TRUCK PACKOLOGY MINI MANUAL

The Real Trade Behind The Game

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FROM THE AV EDUCATE TEAM

Why We Made This Booklet

For a lot of people, the truck is where their first real step into the AV industry begins.

You learn quickly that almost every show starts there and ends there. The gear has to come off safely, get where it needs to go, and at the end of the night somehow make its way back onto the truck in a way that still makes sense.

We created Truck Packology to help make that process easier to understand.

The goal is simple: help you build safer packs, work faster with your crew, and show up better prepared for load-ins and load-outs. Not by overcomplicating it, but by teaching the logic behind the pack in a way that makes sense when you finally see it in the real world.

Our team has put countless hours into turning Truck Packology into a practical and reliable training resource for the AV community. The more people understand how a truck is packed, why things go where they go, and how the crew works together, the smoother and safer the entire process becomes.

THE BOOK TRAINS THE MIND

Learn what the gear is, where it belongs, and the logic of a good pack.

LOAD-INS AND LOAD-OUTS TRAIN THE BODY

Real trucks. Real weight. Real crews. Nothing replaces getting the hands-on experience.

THE GAME GIVES YOU THE REPS

The game is meant to be fun, but the repetition is where the learning happens. The more packs you work through, the more you start to recognize the patterns, see the problems, and understand what makes a load work.

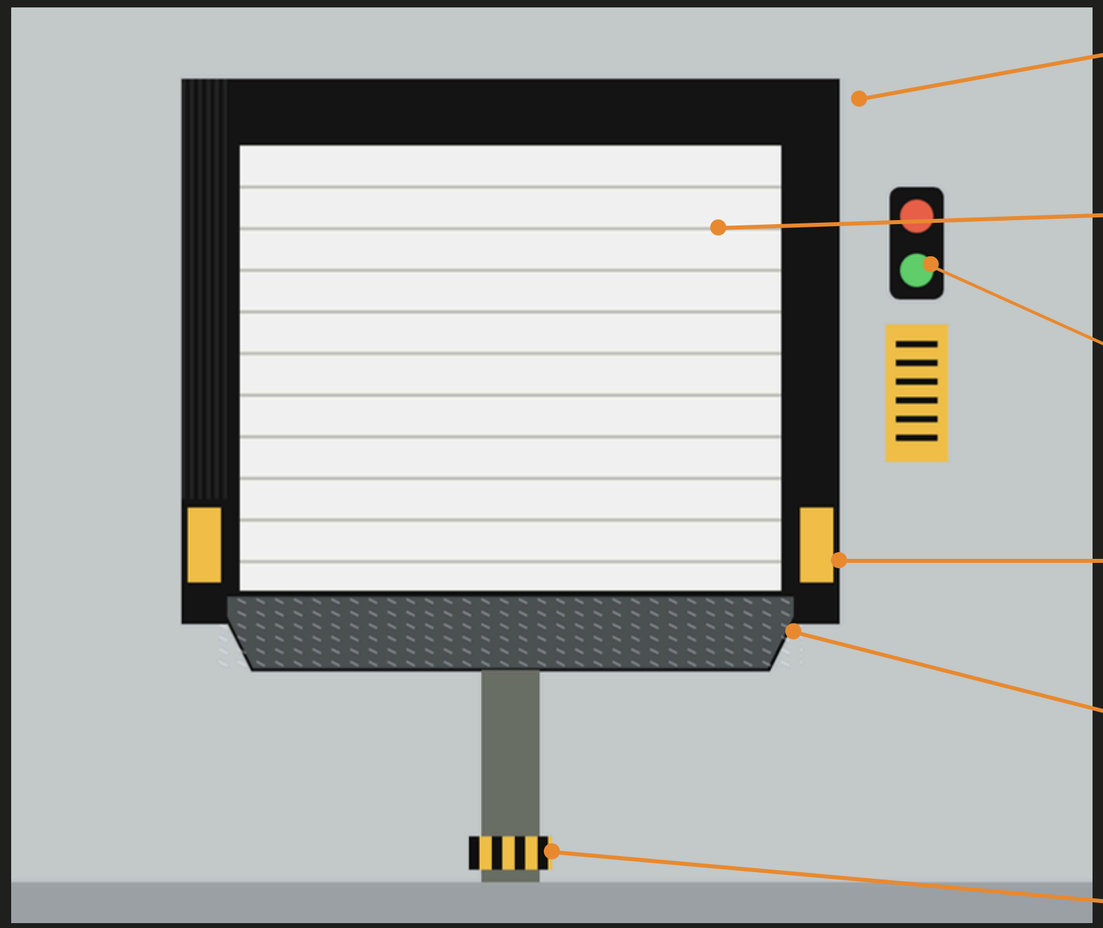
Learn it here. Practice it here. Then take it back to the field.

— The AV Educate Team

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THE LOADING DOCK

Every part of a dock, and what it's for



DOCK SEALS / SHELTER

Padded curtains that seal around the back of the truck.

DOCK DOOR

The roll-up door into the building.

LIGHTS

Red = stop, green = go, for the driver and the crew.

DOCK BUMPERS

The truck backs into these. That bump pushes the load toward the doors.

DOCK LEVELER

A hinged plate that bridges the dock to the truck bed.

VEHICLE RESTRAINT

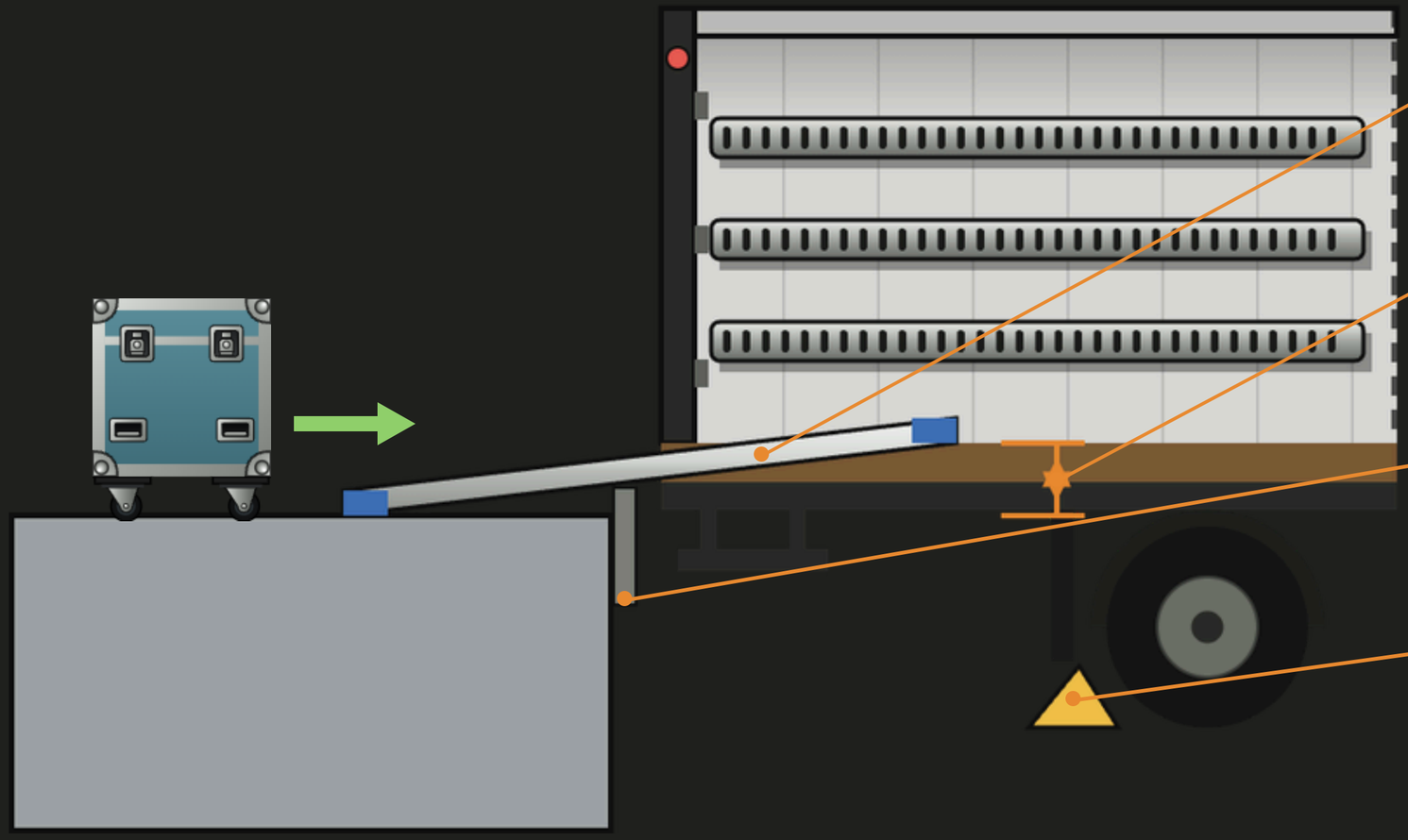
Hooks the truck's rear bumper so it can't pull away while the crew works.

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THE DOCK PLATE

It bridges the gap and the step between the dock and the truck

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DIAMOND PLATE

Aluminum with a grip pattern. The colored edges show where it ends.

HEIGHT DIFFERENCE

The truck bed sits higher or lower than the dock. Each plate has a safe range.

SAFETY LEG

Drops into the gap so the plate can't slide out.

WHEELS CHOCKED

Before anyone crosses: truck tight to the dock, engine off, wheels chocked.

HAND TRAFFIC ONLY

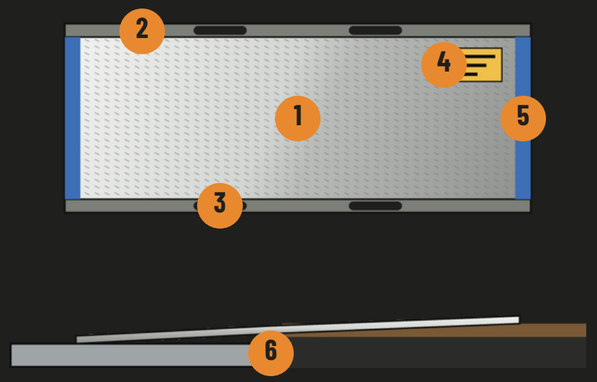
Carts, hand trucks, manual pallet jacks and rolled road cases. Never a forklift.

DOCK PLATE VS DOCK LEVELER

Two ways to bridge the dock and the truck. You'll run into both.

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PORTABLE DOCK PLATE



1 · Diamond-plate deck
2 · Curbs (raised side edges)
3 · Hand holds
4 · Capacity tag
5 · Colored ends
6 · Locking legs

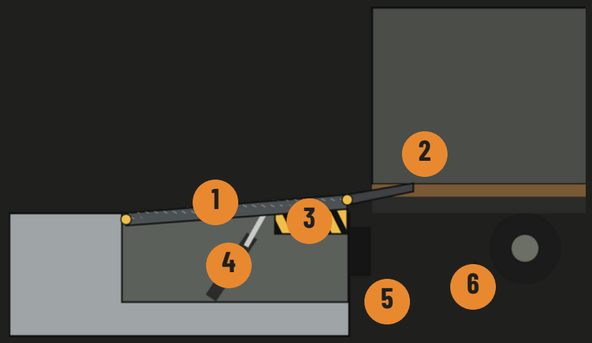
SPECS & LIMITS

- Aluminum, light enough for two people to carry
- Rated load is on the tag: light-duty plates hold a few thousand lb
- Bridges only a few inches of height: stay inside its range
- Hand traffic only: carts, hand trucks, rolled cases. Never a forklift.

CHECK BEFORE YOU USE IT

- The tag's rating and height range fit the job
- No cracks, bends or loose legs
- Legs down in the gap, both ends flat and firm
- Truck chocked or restrained before anyone crosses

BUILT-IN DOCK LEVELER



1 · Deck (hinged ramp)
2 · Lip (rests on the bed)
3 · Toe guards
4 · Hydraulic cylinder
5 · Bumpers
6 · Wheel chock

SPECS & LIMITS

- Built into the dock: often about 6–8 ft long and 6–7 ft wide
- Rated for forklifts: often 25,000–40,000 lb, check its tag
- Works about 12 in above or below dock level
- Push button (hydraulic) or pull chain (mechanical)

CHECK BEFORE YOU USE IT

- Truck chocked or restrained, dock light on green
- Lip lying flat on the bed, not on its edge
- Deck not too steep for your cart or pallet jack
- Stand clear while it moves; store it level when done

USE A DOCK PLATE IN 6 STEPS

For hand traffic: carts, hand trucks and rolled road cases

1. **TRUCK SET**
   Backed tight to the dock, engine off, wheels chocked (or the restraint on).

2. **INSPECT**
   No cracks, bends or loose handles. The dock is dry and clear of trip hazards.

3. **CHECK THE FIT**
   The step up to the truck bed is within the plate's safe range.

4. **LIFT IT RIGHT**
   Use the handles, knees bent, back straight. Heavy plate? Two people.

5. **SET IT DOWN**
   Safety leg down into the gap, both ends flat and firm on the dock and the bed.

6. **CROSS, THEN STORE**
   Roll across smoothly; don't stop or turn on it. Then lift it out and put it away.

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DOCK PLATE RULES

What never to do, how AV crews work it, and what the game skips

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NEVER

Drive a forklift or a powered pallet jack over a portable dock plate. That takes a heavy-duty dock board.

Overload it. Check the plate's rating.

Leave it where people walk.

AV CREW PRACTICE

Take extra care on dock plates and ramps.

Going down a ramp, a pusher controls the pace. Never let a case free-wheel down for someone to catch.

Double-stacked cases: always two people.

(“Gear is replaceable you are not”)

IN THE GAME

The truck backs in with its beeper and the cases roll straight in: the game skips the dock plate.

The DOCK STACK is built "on the dock".

The ACCELERATING phase pushes the load toward the doors, like a dock bump does.

TRUCK SIZES: HOW LONG

Every truck in the game, plus what you'll see on real shows, to scale



10 FT BOX TRUCK

10 ft · 120" · game



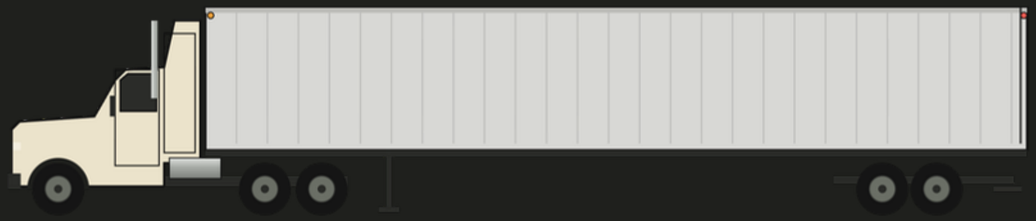
12½ FT BOX TRUCK

12½ ft · 150" · game



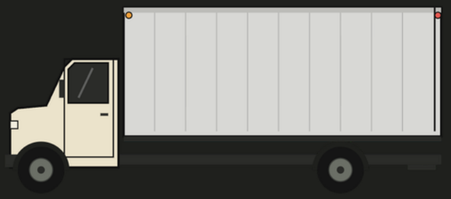
16 FT BOX TRUCK

16 ft · 192" · game



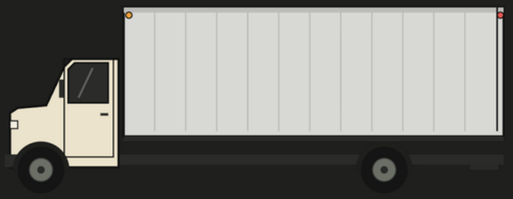
53 FT TRAILER

about 52' 6" (630") inside · real world



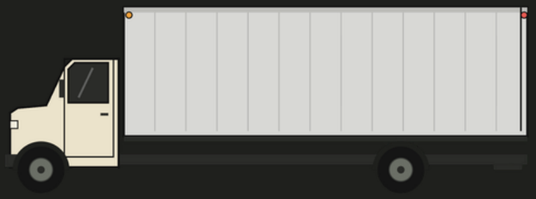
20 FT BOX TRUCK

20 ft · 240" · game



24 FT BOX TRUCK

24 ft · 288" · game and Classic



26 FT BOX TRUCK

about 25–26 ft inside · real world

PRACTICE BAY

6 ft · 72" · game only, not a truck

Everything in the game fits in a box truck. On real shows you'll also see 26 ft box trucks and 53 ft semi trailers: same rules, more rows.

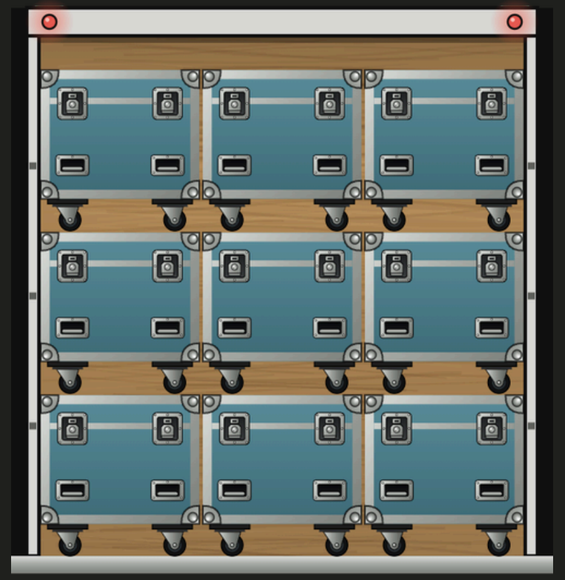
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TRUCK SIZES: HOW WIDE AND TALL

Looking in from the doors, all to the same scale

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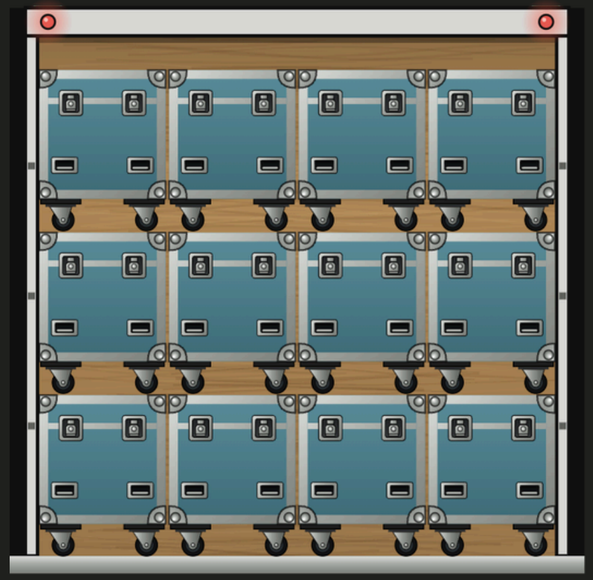
90" BOX TRUCK



90" wide × 96" tall inside

game · some real box trucks

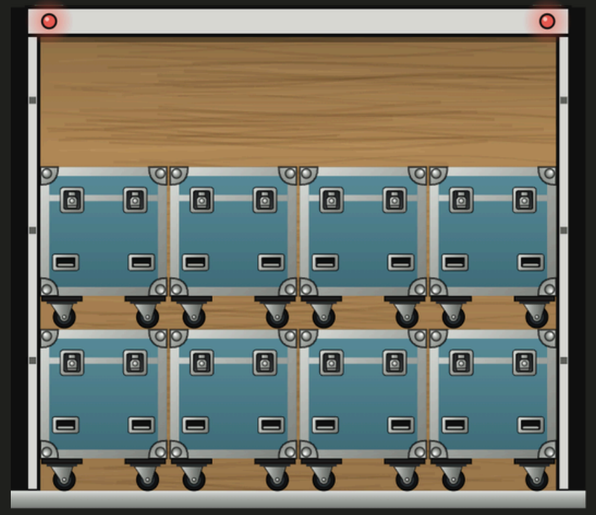
96" BOX TRUCK



96" wide × 96" tall inside

game · most real box trucks

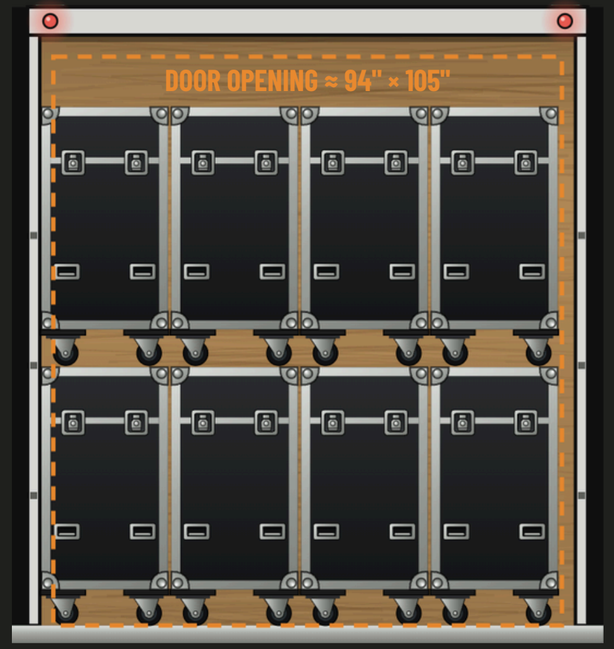
PRACTICE BAY



96" wide × 84" tall inside

game only

53 FT TRAILER



about 99" wide × 109" tall inside

real world · 98–100" × 108–110"

MEASURE EVERY TRUCK

Real trucks change from maker to maker: the sizes are a guide, the tape measure is the truth

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BEFORE YOU PLAN THE PACK, MEASURE

LENGTH on the floor, nose wall to the doors.

WIDTH between the E-track rails, not wall to wall: rails and liners eat inches.

HEIGHT inside, and the DOOR OPENING. It's narrower and shorter than the inside, so a stack that fits inside can still hit the door frame.

Anything in the way: wheel wells, a lift gate, a ramp.

IN THE GAME

Box trucks from 6 to 24 ft long.

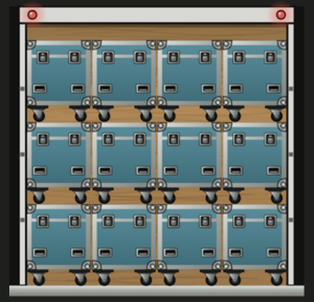
96" wide, and a few 90" trucks that change the math.

96" tall inside when you stack (the Practice Bay is 84").

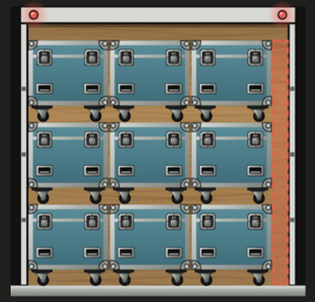
The level card always shows the truck: read it before the first case.

TRUCK MATH 1: ACROSS

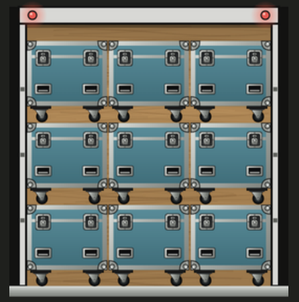
Truck width ÷ case side = cases across. What's left over is the gap.



✓ 96" ÷ 24" = 4 across · 0" left



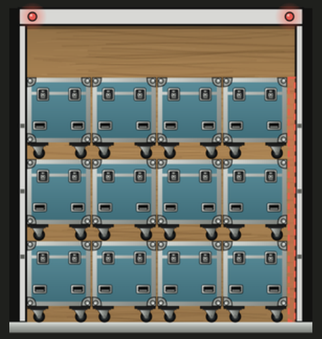
✗ 96" ÷ 30" = 3 across · 6" left: it can slide



✓ 90" ÷ 30" = 3 across · 0" left (turn them!)



✗ 90" ÷ 24" = 3 across · 18" hole



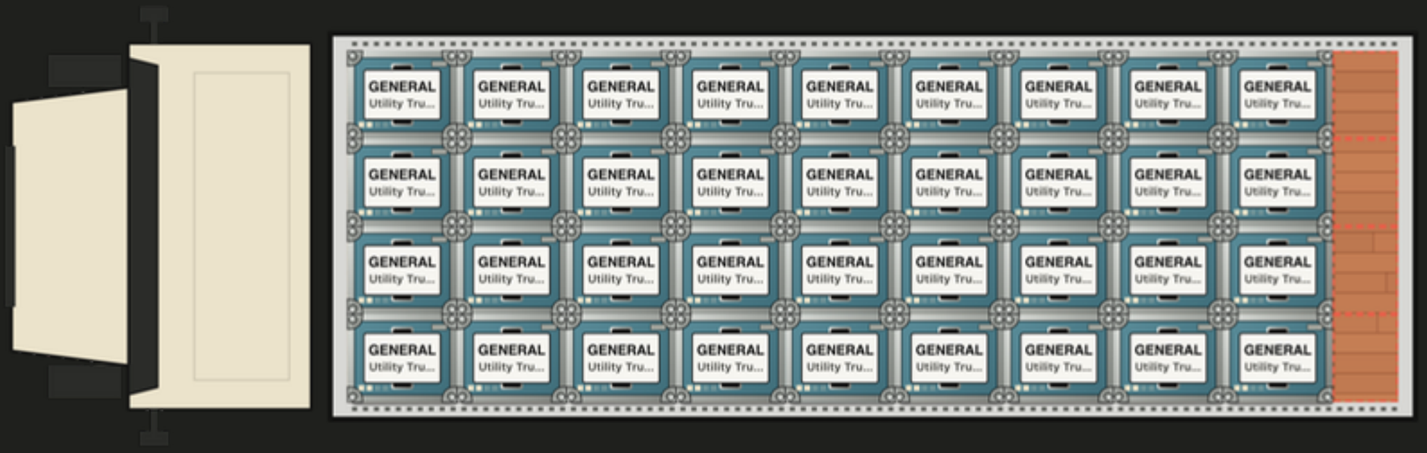
✓ 99" trailer ÷ 24" = 4 across · 3" left

Road cases are built on 24" and 30" sides: turn a case and its sides swap. In the game a gap of 6" or more lets a case slide. On a real truck, close it: turn cases, mix sizes, or tie it in.

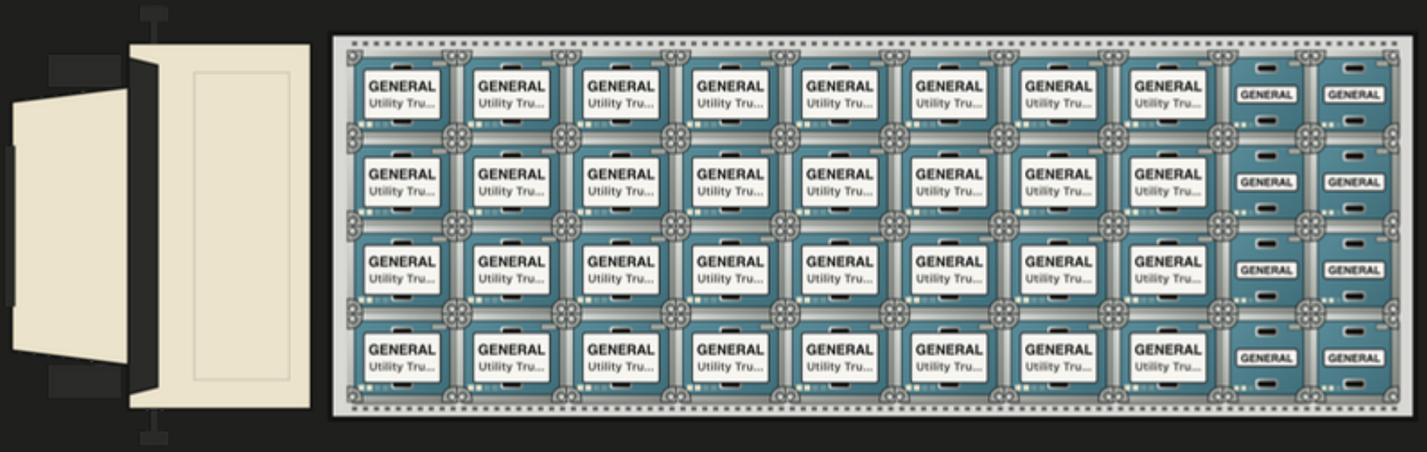
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TRUCK MATH 2: DOWN

Truck length ÷ case depth = rows. A 24 ft box truck from above, nose (cab) on the left.



✗ 24 ft = 288" ÷ 30" = 9 rows · 18" left at the doors

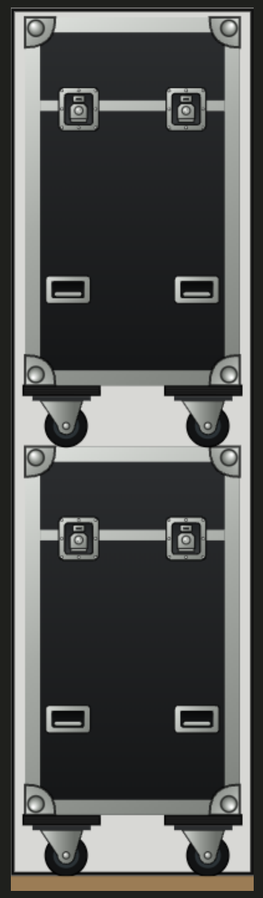


✓ 8 rows of 30" + 2 rows turned to 24" = 288" exactly

Same idea down the truck: rows of 30" leave 18" at the doors of a 24 ft truck. Turn two rows so they're 24" deep and the truck is full to the doors.

TRUCK MATH 3: UP

Inside height ÷ case height = how high you can stack. A 96" truck, from the doors.



✓ 48" + 48" = 96"



✓ 30" × 3 = 90"



✓ 48" + 30" = 78"

ADD IT UP BEFORE YOU LIFT

96" truck: 48" + 48" fills it, or three 30" cases make 90".

A real trailer may be 108–110" inside, but check the door opening too (about 105").

Heavy goes low, and the stack has to come off in order.

On the truck, set only one case on top of another within reach. Build anything taller on the dock first.

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KNOW THE LIFT LIMIT

Max is the most anyone should lift in perfect conditions. The safer target is where a real crew should stay.

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LIFT	MAX · IDEAL CONDITIONS	SAFER TARGET
1 PERSON	51 lb · 23 kg	35 lb · 16 kg
2 PEOPLE	87 lb · 39 kg	68 lb · 31 kg
3 PEOPLE	107 lb · 49 kg	77 lb · 35 kg
4 PEOPLE	One on each corner, one voice. Too heavy for four? Use the forklift.	

WHY A TEAM LIFTS LESS

Every extra person adds less than their own limit. Nobody lifts in perfect sync, and one person ends up with more of the weight.

THE LIMIT DROPS WHEN YOU...

- Reach out: arms 17 in from the body cuts it about 40%
- Twist: turning your back while you lift
- Repeat: case after case, hour after hour
- Work a slope or step: the low end carries more

ON A REAL TRUCK: most road cases weigh more than any hand-lift limit. Roll them on their casters and only lift to stack, and only what your team can lift. Over 70 lb again and again? Add a mechanical aid, not more hands.

Guidelines, not legal limits: OSHA sets no single weight limit. Max = Revised NIOSH Lifting Equation (51 lb) with the EN 1005-2 team factor. Safer target = UK HSE team-handling guidance (two people 2/3, three people 1/2 of their combined limit).

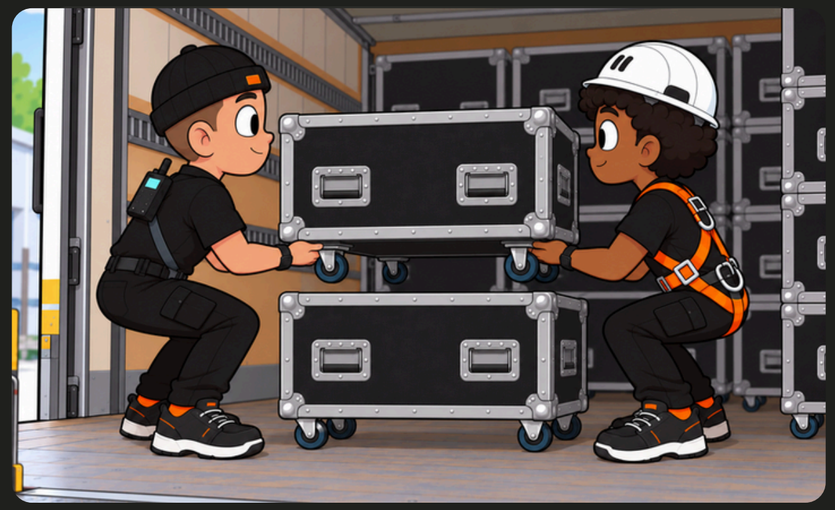
SAFE STACKING

Lift together, stop at the shoulder, and let the gear take the heavy ones

Illustrations from our comic book, Truck Packology: made for kids, fun for all ages.



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1 · CALL THE LIFT

One voice counts it. Two crew lift together, knees bent, case close, and set it into the caster cups below.



2 · THE FOUR-PERSON LIFT

Big cases get one person on each corner. Same rules as two: one voice calls it, lift together, set down together. Four share the weight, but never perfectly evenly.



3 · TOO HEAVY? USE THE GEAR

Let a machine do the lift: a forklift with a trained driver, a lift table or a pallet jack. The crew stands clear.

HEIGHT RULES

By hand, stack no higher than the shoulder (60–65 in). A free-standing stack stays under 3× its base width unless it's strapped in.

ONE VOICE

One person calls every team lift. Lift together, set down together, and say it out loud when you need to stop.

IN THE GAME

Stacking opens at level 31. Match cases, heavy on the ground and light on top and every stacked case gets tied in.

NEW CASES ON SITE? 3 STEPS

Case sizes change from show to show. The math doesn't.

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1 MEASURE THE CASE

Width × depth × height, and which way it has to face (wheels down, "this side up").

2 DO THE THREE DIVISIONS

Truck width ÷ case side (try both sides).
Truck length ÷ case depth = rows.
Inside height ÷ case height = how high.

3 FIX THE LEFTOVER

Turn cases, mix sizes (24 + 30 + …), or give the odd cases a row of their own. A gap you can't close gets a bar or a strap.

QUICK CHECK

96" truck: four 24s, or three 30s + a 6" gap.

90" truck: three 30s.

24 ft truck: 9 rows of 30" + 18", or 8 × 30" + 2 × 24".

96" tall: 48 + 48, or 30 + 30 + 30.

RATCHET STRAPS AND E-TRACK

The parts, and how a strap holds on to the truck

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E-TRACK ON THE WALL

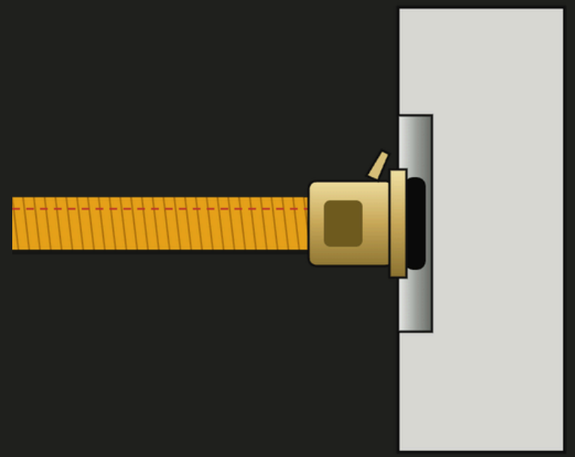


Steel rails bolted to both side walls, in rows up the wall. Straps and load bars clip into the slots, so they can go anywhere.

THE RATCHET STRAP



Two fittings, a length of 2" webbing and a ratchet. Work the handle and the strap winds onto the spool and pulls tight. WLL = working load limit: the most it should hold.



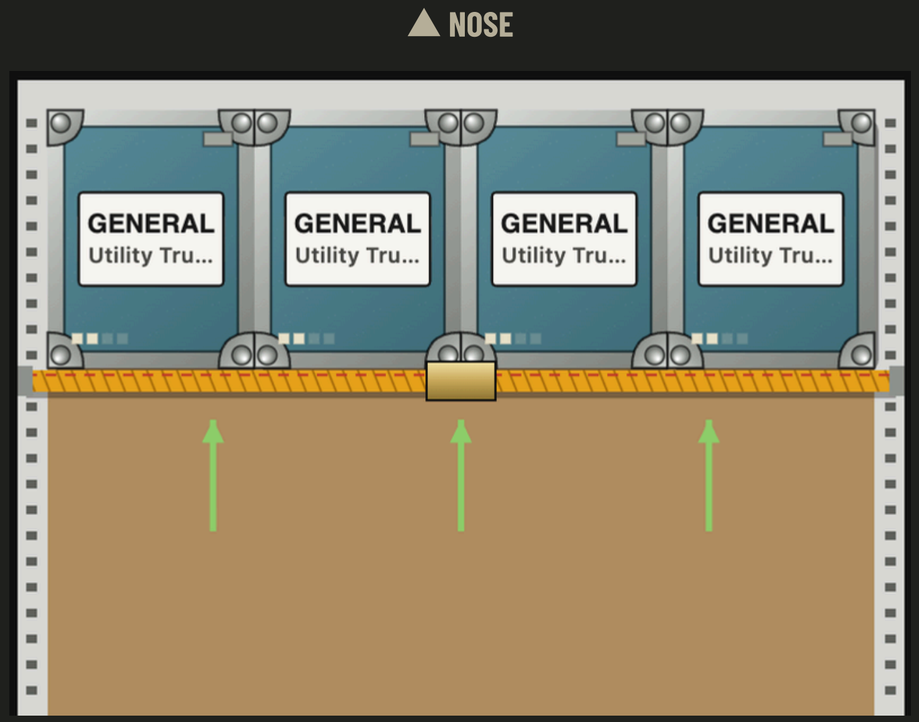
CLIPPED IN

Push the fitting into any slot until the latch clicks. The hooks take the load; the latch stops it popping out. Press the latch to release.

TIED IN: WHERE THE STRAP GOES

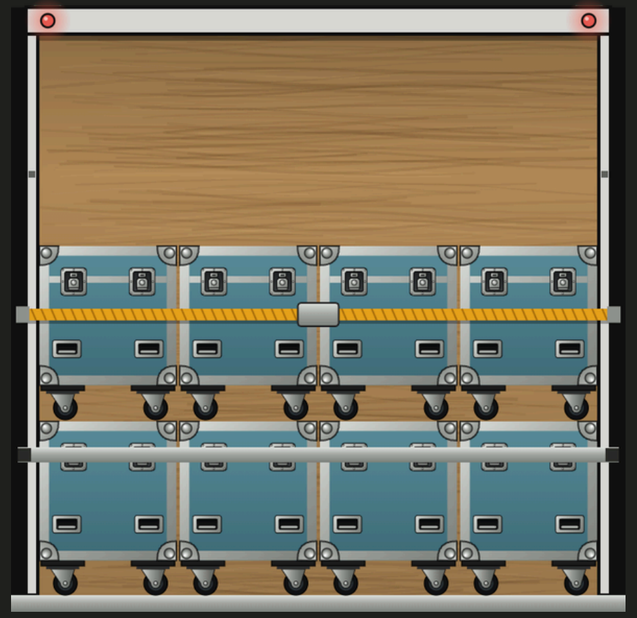
Wall to wall, right behind the tier, at the right height

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TOP VIEW

The strap runs from one wall to the other, right behind the tier, and pulls it toward the nose. Wall to wall, it counts at its full rating.



STRAP
48" RAIL · TOP CASES

LOAD BAR
24" RAIL · FLOOR TIER

DOOR VIEW

Stacked tiers are tied in at two heights, so the top cases can't tip back and the floor tier can't slide.

STRAPS: REAL TRUCK VS THE GAME

What a crew really does, and what the game keeps simple

ON A REAL TRUCK

- AV best practice: strap every 2 - 3 rows as you go, before the next rows go in.
- Wall to wall, at the right height. No knots, no cut or damaged straps.
- The straps together must be rated for at least half the weight they hold (FMCSA).
- §393.110: the number of straps depends on each item's length and weight. A blocked case of 5 ft or less needs one.
- A heavy tier can need two straps or more. The weakest part sets the rating.

IN THE GAME

- One strap holds any tier, whatever it weighs.
- It goes wall to wall, right behind a finished tier.
- Stacked tiers get two: one high, one low.
- Straps never knot, wear out or run out of slots.

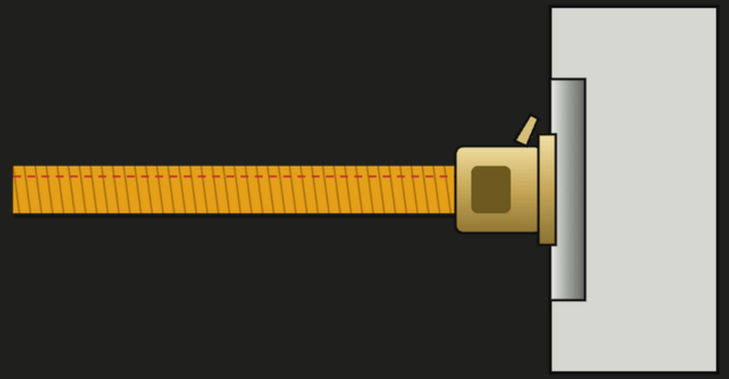
WHY THE GAME KEEPS IT SIMPLE: one strap per tier teaches WHERE and WHEN to tie in. On a real truck, also check the weight and the rating.

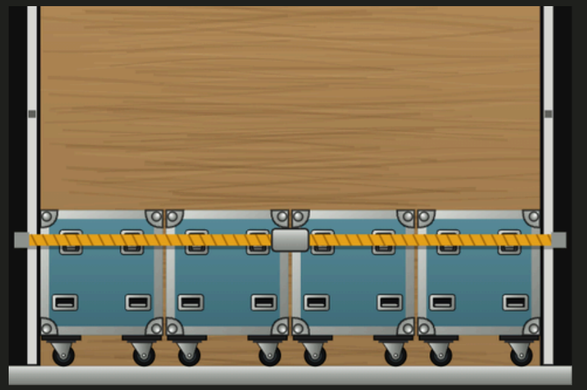
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HOW TO USE A RATCHET STRAP

Steps 1 to 3: set it up and clip it in







1 OPEN IT UP

Hold the release latch and lay the handle all the way flat. Pull the loose end through so the strap is slack.

2 CLIP THE FIRST END

Press the spring latch on the fixed end and push the fitting into a slot on one wall, at the height of the tier. Let go: it clicks in.

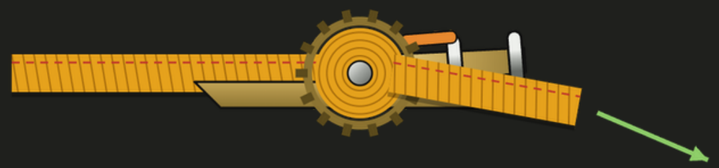
3 ACROSS TO THE WALL

Run the strap flat across the face of the tier, no twists, and clip the other end into the opposite wall at the same height.

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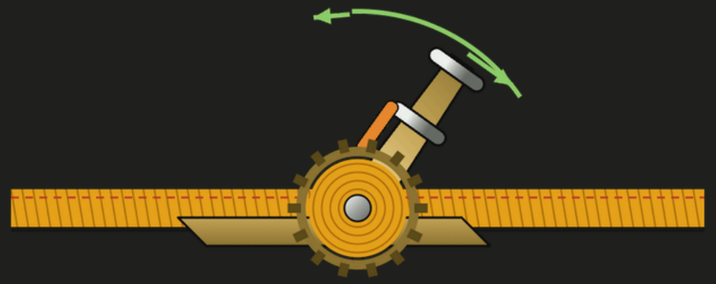
HOW TO USE A RATCHET STRAP

Steps 4 to 6: tighten it and tidy up



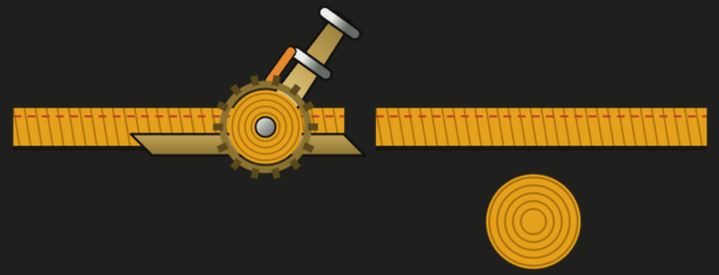
4 PULL THE SLACK

Feed the loose end through the slot in the spool and pull most of the slack through by hand. Too much left to wind up can jam the ratchet.



5 RATCHET IT TIGHT

Pump the handle until the strap is snug: 2 to 4 wraps on the spool. Stop when you can just slide a finger under it. Don't crush the cases.



6 TIDY THE TAIL

Roll up the leftover webbing and tuck it under the strap, so nothing hangs loose or gets caught.

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STRAPS: OFF, CHECKS AND THE LAW

Unloading, looking after your straps, and what the rules say

TAKING IT OFF

1. Pull the release latch and open the handle all the way flat.
2. The strap lets go: pull the webbing out of the spool.
3. Press each fitting's latch and lift it out of the rail.

Keep your face and hands clear when it lets go: a tight strap can snap back.

CHECK IT FIRST

- No cuts, fraying or burns in the webbing.
- The WLL tag is there and you can read it.
- Fittings and ratchet not bent, cracked or rusted shut.
- Damaged? Take it out of use.

THE LAW (FMCSA)

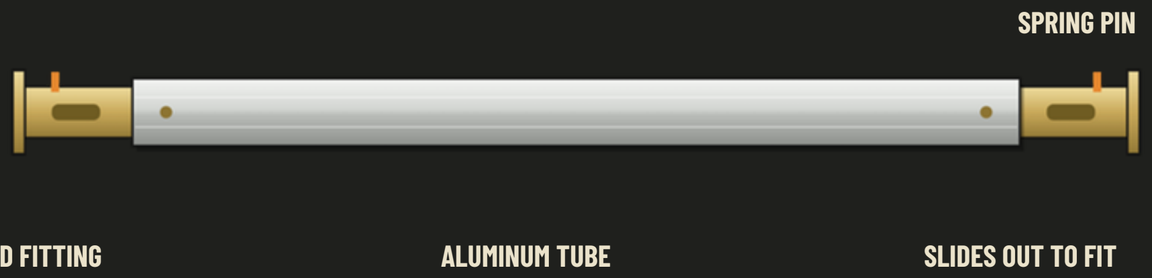
- All the straps together rated for at least half the cargo's weight.
- The weakest part sets the rating: strap, fitting or rail.
- No knots, no damaged straps.
- Pad sharp edges so the strap can't be cut.

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LOAD BARS: TWO KINDS

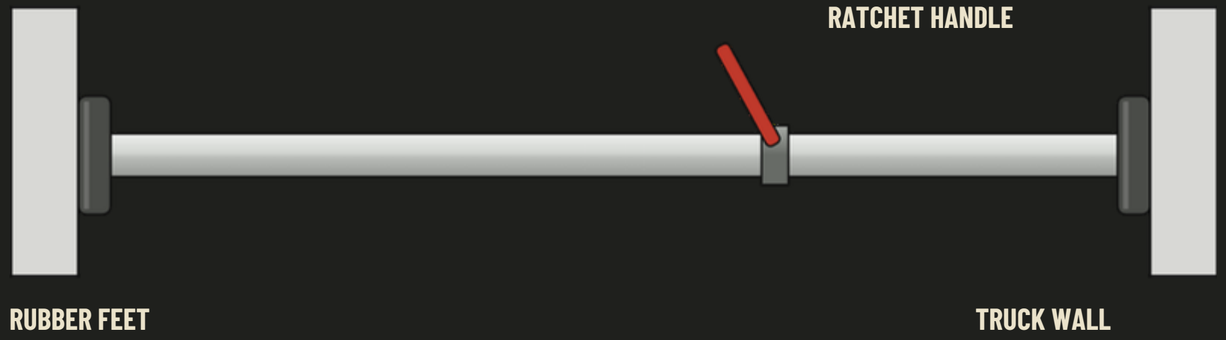
One locks into the E-track. The other just presses on the walls.

E-TRACK LOAD BAR (SHORING BEAM)



- Spring-loaded ends lock into the E-track slots, so it can't pop out easily.
- Typical working capacity 2,200–3,000 lb. Check the tag.
- Aluminum beams weigh about 12–16 lb.
- This is the game's LOAD BAR.

LOAD LOCK (FRICTION BAR)

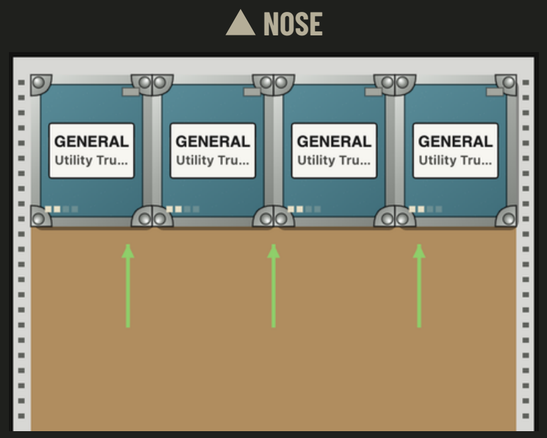


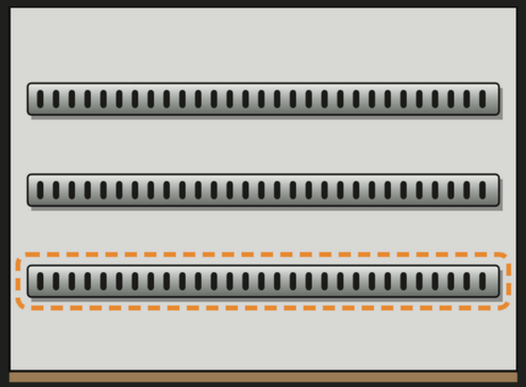
- No track needed: rubber feet press on the bare walls.
- Usually no rating: it holds by friction and how stiff the walls are.
- It can slip if the walls flex.
- Not in the game.

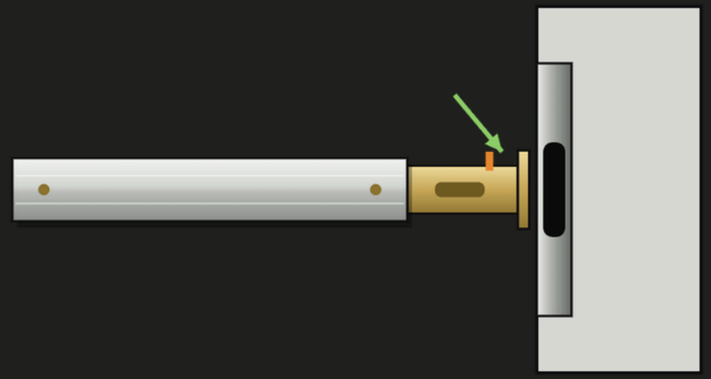
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HOW TO USE A LOAD BAR

Steps 1 to 3: an E-track bar, set up and hooked in







1 GET THE TIER TIGHT

Push the finished tier all the way to the nose, wall to wall, with no gaps. The bar holds a tier in place; it doesn't fill holes.

2 PICK THE HEIGHT

The first bar goes low, on the 24" rail across the floor tier. A second bar (or a strap) goes higher. Same slot height on both walls.

3 HOOK ONE END IN

Pull the spring pin, set the end fitting into a slot on one wall and let the pin snap in. The plate hooks behind the track.

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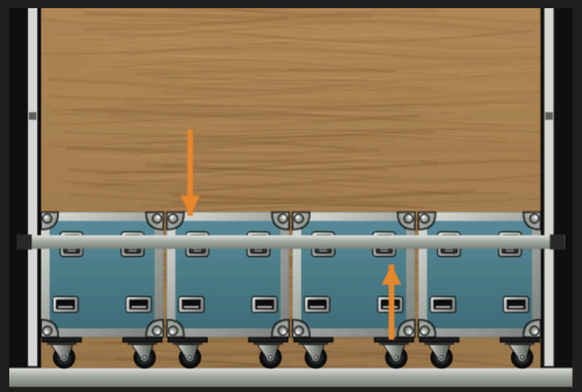
HOW TO USE A LOAD BAR

Steps 4 to 6: across, checked, and back out



4 STRETCH IT ACROSS

Slide the telescoping end out to the other wall and lock it into the same row of slots, so the bar sits level and right against the cases.



5 CHECK IT

Push and pull on the bar. It shouldn't lift out or slide. Added a second bar? Re-check the first: the walls flex and can loosen it.



6 TAKING IT OUT

Keep hold of the bar. Pull the spring pin, lift one end out of the track, then the other. A loose bar can drop or swing.

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LOAD BARS: REAL TRUCK VS THE GAME

Using a load lock, what crews do, and what the game keeps simple

USING A LOAD LOCK

1. Set it 1–2" shorter than the gap.
2. Feet flat and square on both walls, right behind the cargo.
3. Pump the handle until the feet bite, about 6 clicks. More can bow the walls.

To release it, keep hold of the handle: it can snap back.

ON A REAL TRUCK

- A bar holds a tier in place. It doesn't fill holes.
- Two friction bars: don't crank the first too hard. Even out the pressure, then re-check both.
- Check the rating, and that the bar spans the truck's width.
- Bars block a row; straps hold single or fragile cases.
- Store bars in a rack, off the floor.

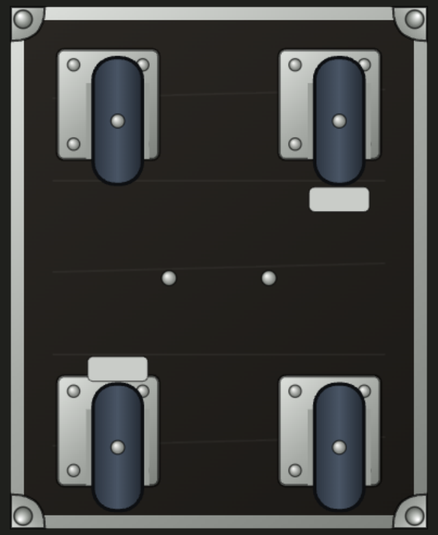
IN THE GAME

- The LOAD BAR is an E-track bar: wall to wall, on the 24" rail.
- Drop it right behind a finished tier.
- The nose tier needs a bar before anything is stacked on it.
- It never slips and you never run out of slots. On a truck, check it every time.

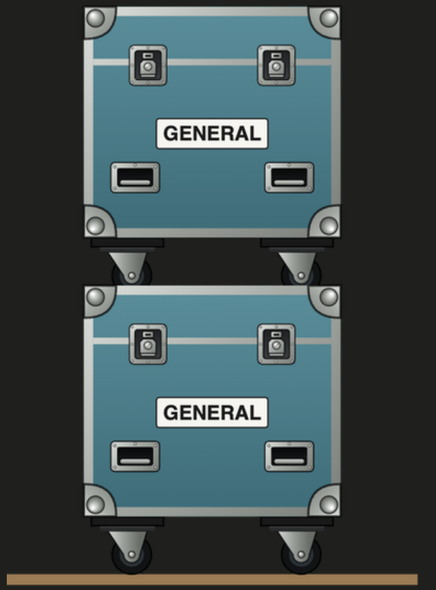
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CASTER CUPS: WHY CASES STACK

A road case on wheels can ride on another case, when the casters have somewhere to sit







1 THE BOTTOM

Four swivel casters on bolted plates, near the corners. Two of them have brakes.

2 THE LID

Four caster cups: steel dishes set into the lid, right where the casters of the case above land.

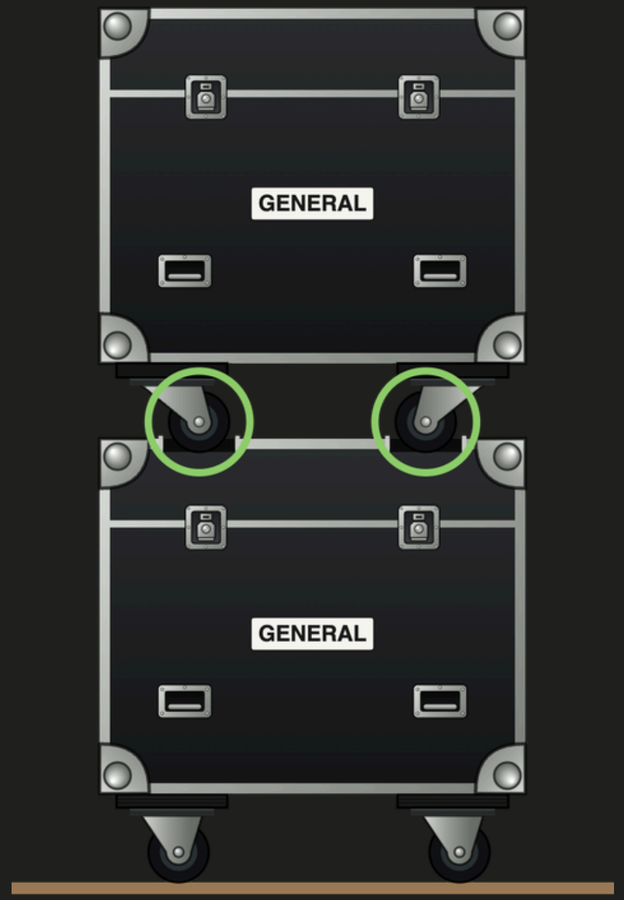
3 STACKED

The casters drop into the cups, so the top case can't roll off its stack.

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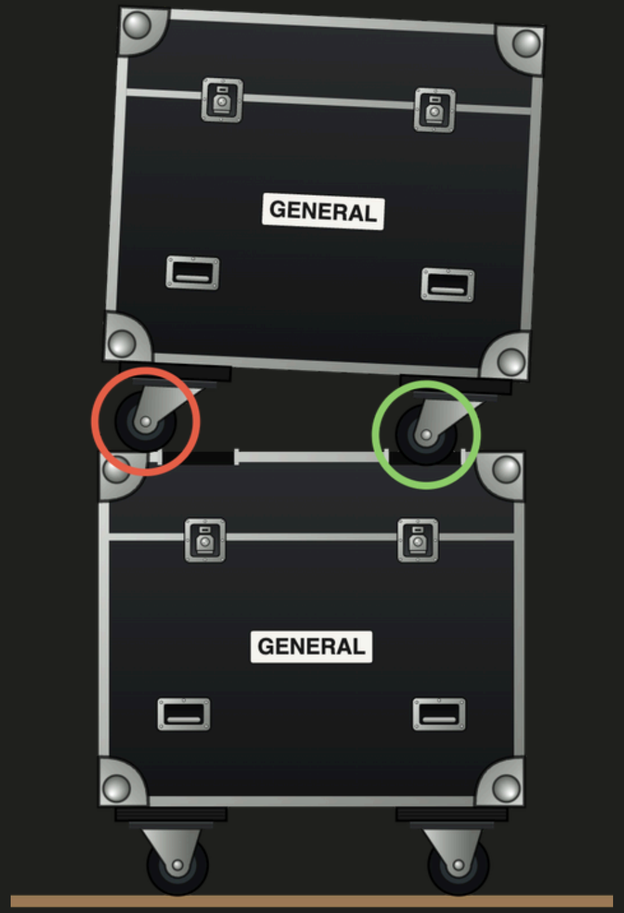
SIDE VIEW: INTO THE CUPS

Wheels matched to the cups drop in. One wheel turned the wrong way and the case won't line up.



✓ BOTH WHEELS IN THE CUPS

Both wheels turned to match the cups. They drop in and the top case sits level and can't roll off.



✗ LEFT WHEEL FACING OUT

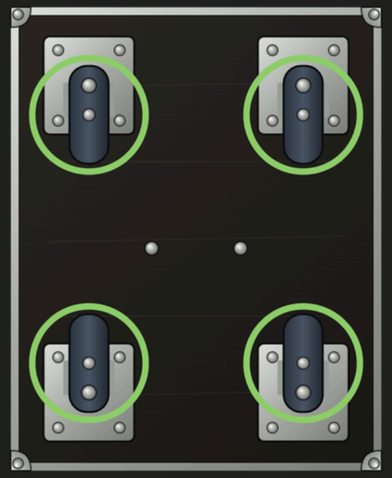
It misses its cup and rides up on the lid. The case sits crooked and can roll off the stack.

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MATCH THE WHEELS

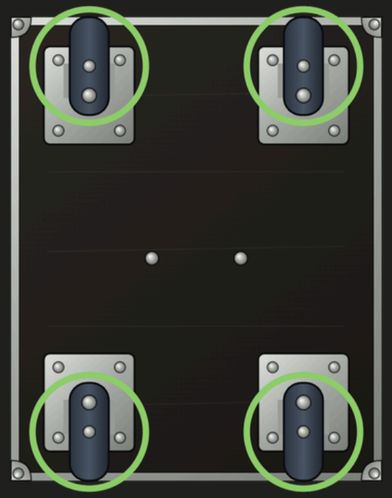
All four wheels the same way, all in or all out, to match the cups. The bottom of the case, flipped over.

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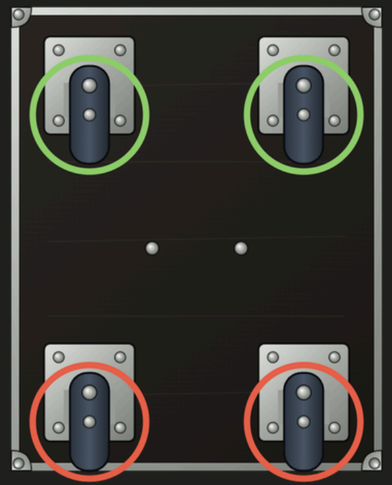
✓ ALL TUCKED IN

Every wheel turned toward the middle. Each one drops into its cup.



✓ ALL TUCKED OUT

Every wheel turned toward the edges. Right when the case's cups are set wide.



✗ ONE ROW OUT

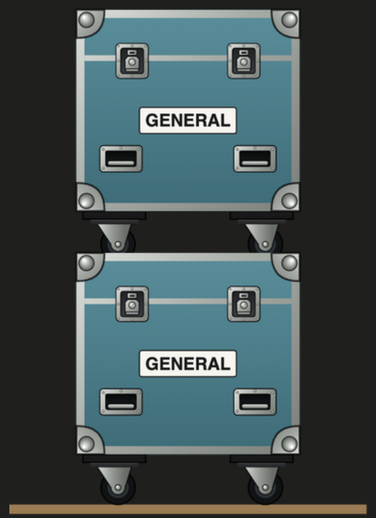
One row tucked in, one row tucked out. They can't all line up with the cups, so the case sits crooked and can roll.

WHY IT MATTERS

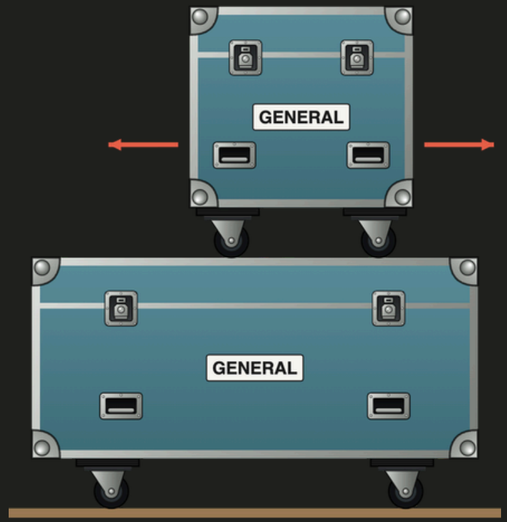
A swivel wheel trails behind its pivot, so where it lands depends on which way it's turned. Some cases are cupped for wheels in, some for wheels out: turn all four to match, then set the case down together.

STACKING: REAL TRUCK VS THE GAME

Matching cases, so the casters land in the cups



✓ SAME FOOTPRINT
Casters in the cups



✗ SMALLER ON BIGGER
Casters miss the cups and can roll

ON A REAL TRUCK

- Stack matching cases: same footprint, so the casters land in the cups.
- Heavy low, fragile high, screens on top.
- Two people on every double-stacked case.
- You only reach over the first row, maybe the second. Deeper stacks are built on the dock first.
- The last cases on wheels: set the brakes or block them so nothing rolls.

IN THE GAME

- Any case can sit on a flat, fully covered lid of the right size.
- The game doesn't check for cups: it trusts you to stack matching cases.
- Casters never roll and nobody has to lift.
- The reach rule matches: over one row, and deeper stacks go up on the dock.

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QUICK GUIDE · RATCHET STRAP

The whole strap on one page



Fitting · 2" webbing · ratchet + handle · WLL tag · fitting

USE IT IN 5 STEPS

1. CLIP IN: press each fitting's latch and push it into a slot on each wall, at the tier's height.
2. THREAD: open the handle flat and feed the loose end through the spool slot.
3. TAKE THE SLACK: pull it tight by hand. Leave little to wind up.
4. RATCHET: pump until snug, 2 to 4 wraps. Don't crush the cases.
5. LOCK IT: handle all the way down. Roll up the tail and tuck it in.

TO RELEASE: hold the release tab, swing the handle all the way open, pull the webbing out. Keep your face and hands clear.

READ THE TAG

- WLL (working load limit) is the most it should ever hold: about a third of the breaking strength.
- 1": light duty, up to about 1,000 lb.
- 2": the truck strap, up to about 3,333 lb.
- 3–4": heavy duty. Straps that look the same can be rated very differently.

THE LAW (FMCSA)

- All the straps together rated for at least half the load's weight.
- The weakest part sets the rating: webbing, ratchet, fitting or track.
- No knots, no cut or damaged straps. Pad sharp edges.

CARE: check for fraying, cuts, burns and mold. Store cool, dry and out of the sun.
IN THE GAME: the strap ratchets itself, and one holds any tier.

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QUICK GUIDE · LOAD BARS

Both kinds, in and out

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E-TRACK LOAD BAR

also: shoring beam, logistic bar · the game's bar

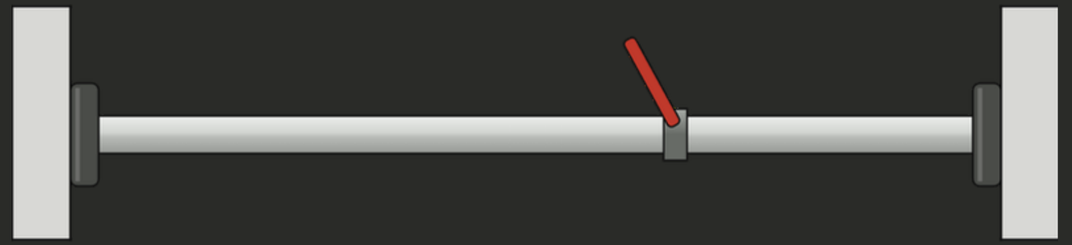


1. Push the tier tight to the nose.
2. Pull the pin, hook one end into a slot.
3. Slide the other end out and lock it at the same height.
4. Push and pull: it mustn't move.

TAKE IT OUT: hold the bar, pull the pin, lift each end out.

FRICTION BAR (LOAD LOCK)

also: cargo bar · not in the game



1. Set it 1–2" shorter than the gap.
2. Feet flat and square on both walls, right behind the cargo.
3. Pump until the feet bite, about 6 clicks.
4. Push and pull: it mustn't move.

TAKE IT OUT: hold the handle firmly. It can snap back.

RIGHT SIZE: check the bar's range against the truck's inside width.
TWO FRICTION BARS: don't crank the first hard. Even them out, re-check both.
BAR + STRAP: a bar blocks a row; straps hold single or fragile cases.

DECKING: E-track beams at one height with plywood on top make a second floor.
STORE THEM: back in the rack, off the floor.
IN THE GAME: the bar snaps onto the 24" rail and never slips.

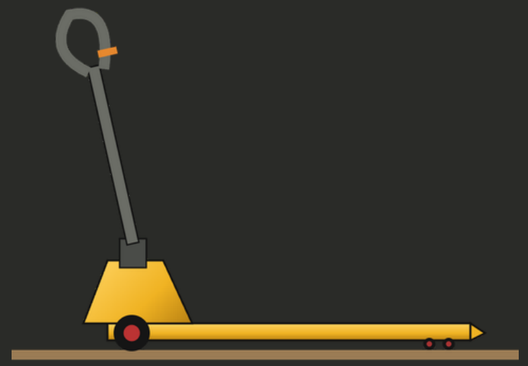
QUICK GUIDE · PALLET JACKS

Manual and electric: the parts and how to use them

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MANUAL

a hand pump lifts, you pull it



CONTROL LEVER
raise / neutral / lower

HANDLE
steer and pump

Up to about 5,500 lb. Lighter loads, tight spaces.

1. Check it: no cracks, leaks or worn wheels.
2. Lever up: forks down. Push them all the way into the pallet, centered.
3. Lever down, pump until it's about 1" up. Lever to neutral.
4. Pull to steer: slow, with wide turns.
5. Stop, lever up to lower, forks out.

ELECTRIC (WALKIE)

a battery motor drives and lifts



BELLY BUTTON
backs off if it touches you

EMERGENCY STOP
the red button

BATTERY GAUGE

Up to about 15,000 lb, for longer runs. It's a powered truck: OSHA 1910.178 says you're trained and evaluated first, and re-checked every 3 years.

1. Charged, unplugged, no damage. Test the horn and the belly button.
2. Forks all the way down, then all the way into the pallet.
3. Lift about 1" with the up button.
4. Drive slowly. Walk beside it or pull it.
5. Stop, feet clear, lower it.

PALLET JACKS: SAFETY EVERY TIME

Plus the dock plate, and where they show up in AV

SAFETY, EVERY TIME

- BOOTS + GLOVES: steel-toed boots and gloves.
- FEET CLEAR: keep your feet out from under the forks and the wheels.
- NEVER RIDE: nobody stands or rides on a pallet jack.
- LOAD LOW: travel with the load just off the floor, about 1".
- SLOPES: straight up or down, never across, load on the uphill side.
- BLIND CORNERS: slow down. Electric: sound the horn at corners and doorways.

ON THE DOCK PLATE

- A manual pallet jack can cross a portable dock plate.
- An electric jack can't: it needs a heavy-duty dock board.
- Check the plate's rating either way.

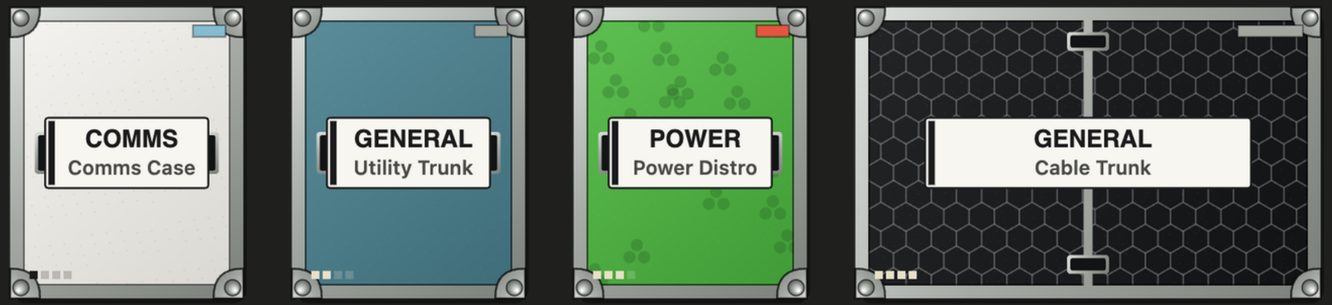
IN AV AND IN THE GAME

- Shows and tours: road cases roll on their own casters.
- Expos: most gear comes wrapped on pallets. Bigger loads on a pallet jack move faster around an expo hall.
- The game has no pallets or pallet jacks.

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WEIGHT PIPS: HEAVY IN THE NOSE

Count the squares on the lid: 1 is light, 4 is very heavy

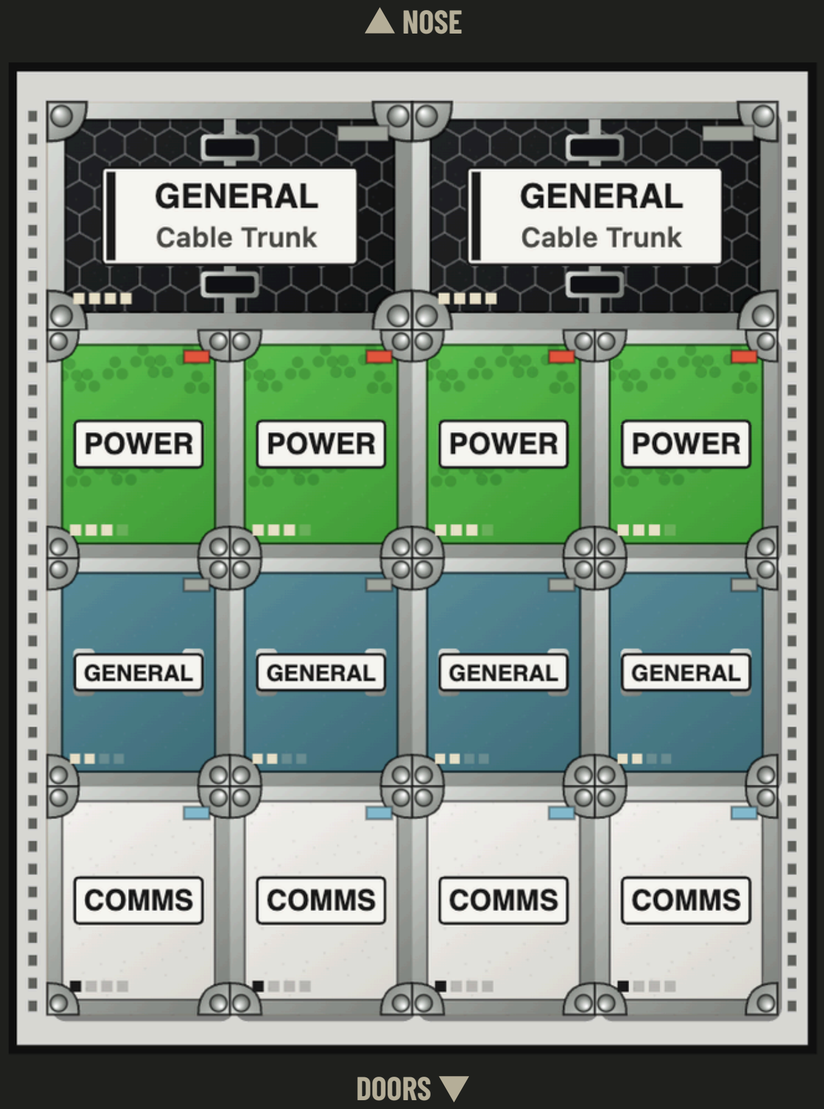


1 · LIGHT **2 · MEDIUM** **3 · HEAVY** **4 · VERY HEAVY**

IN THE GAME: the pips sit bottom left on every lid. Heaviest cases go in first, against the nose, and no case may sit on a lighter one.

ON A REAL TRUCK

- Real cases have no pips: know your gear and read the labels.
- Heavy goes low and in the nose: a firm base for the whole load.
- Never heavy on light, or heavy on fragile.
- Two people on anything heavy.



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THE 24 × 30 FAMILY

Road cases are built on 24" and 30" sides, so they fit together



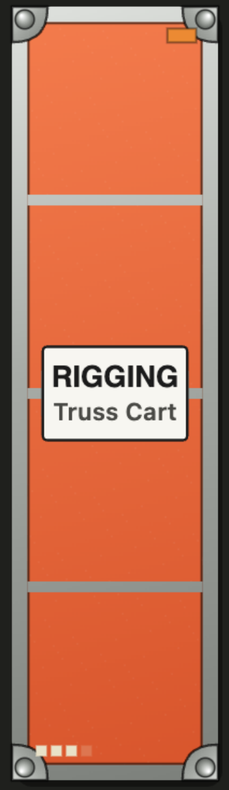
24 × 30



48 × 30 = two 24s



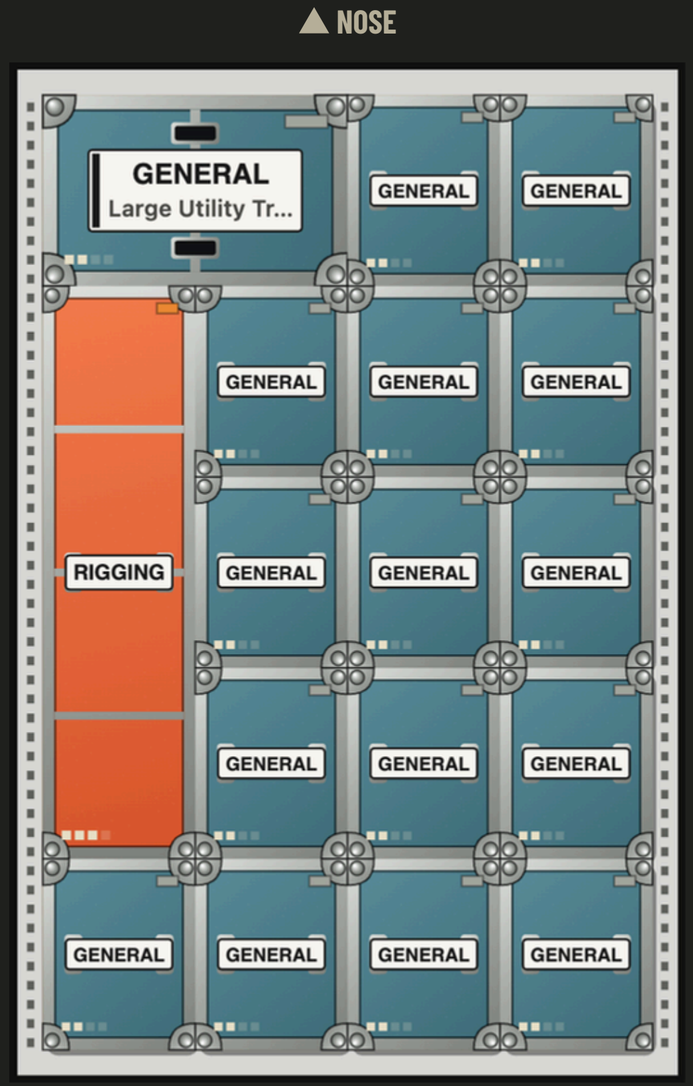
24 × 60
= two 30s



24 × 90
= three 30s

Because every size is built from the same two sides, the family lines up: a 48 sits beside two 24s, a 60 behind two 30s, a 90 beside three.

A 96" truck takes four 24s across, or a 48 and two 24s.

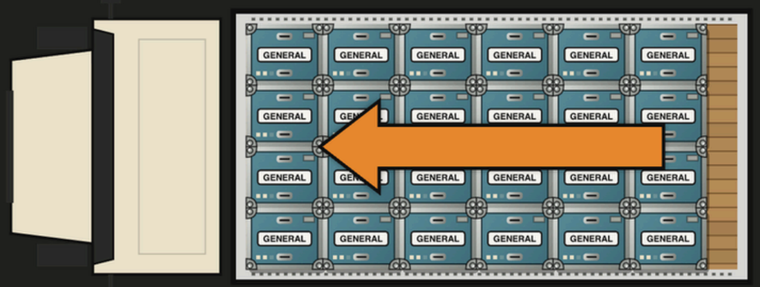


IN THE GAME: everything snaps to a 6" grid. ON A REAL TRUCK: odd sizes and wiggle room, so measure first. Cases off the family (lens cases, Pelicans) leave holes unless you plan a row for them.

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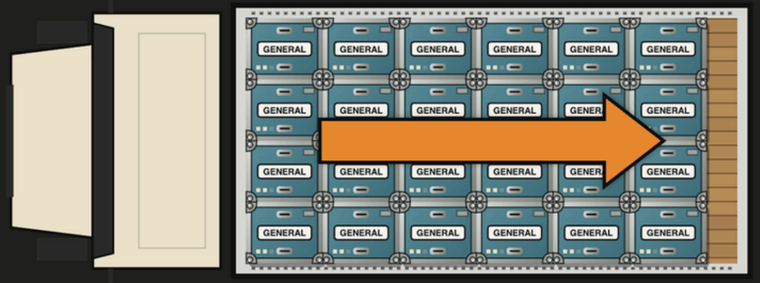
THE THREE FORCES

A loose case moves. Here's what pushes it (FMCSA test forces)



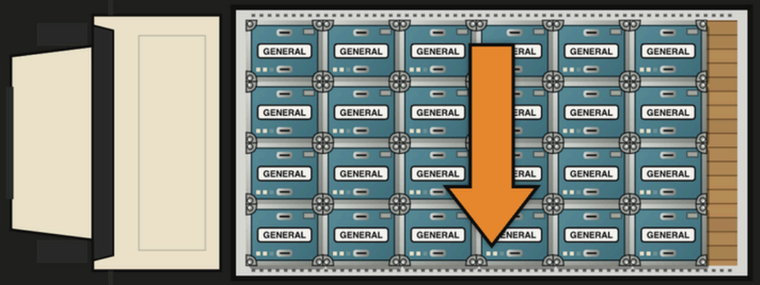
HARD BRAKING · 0.8 g TOWARD THE NOSE

The truck stops, the load keeps going. The biggest of the three.



DOCK BUMP · 0.5 g TOWARD THE DOORS

Backing into the dock and hitting the bumpers. The game calls this push ACCELERATING (pulling away): same direction, different cause.



TURNS · 0.5 g SIDEWAYS

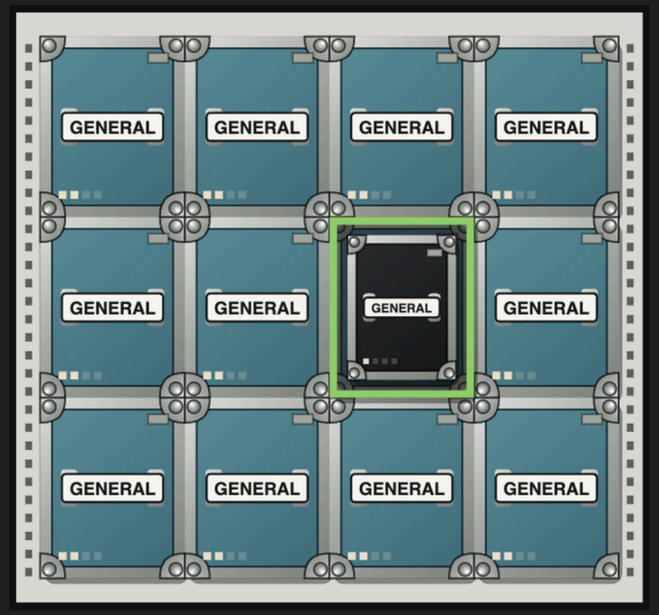
Corners and ramps push the load toward a side wall. A tall, loaded truck can roll over at about 0.35 g.

IN THE GAME: every phase is the same size: any case with room slides, and a fall fails the level. On a real truck the fix is the same: tight, blocked and tied in.

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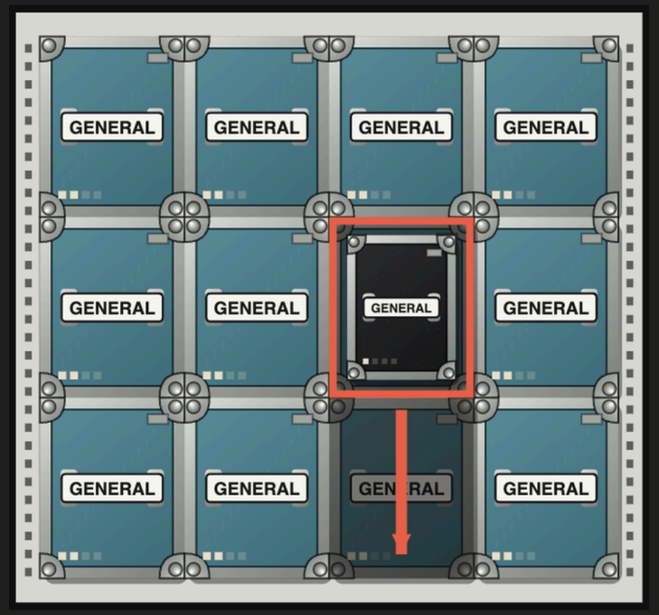
THE POCKET: SMALL HARD CASES

Too small to strap, so they ride boxed in on the lids. From above, nose at the top



✓ BOXED IN

Taller cases on every side. It can shift a little in its pocket, but it can't get out.



✗ ONE SIDE OPEN

Nothing at its height toward the doors, so it slides out.

ON A REAL TRUCK

- Small hard cases (Pelican, Nanuk style) have no wheels and are too small to strap.
- They ride on a lid, with walls, the nose or taller cases at their height on every side.
- You can lift one over the first row, maybe the second. Deeper than that, stack it on the dock first.

IN THE GAME

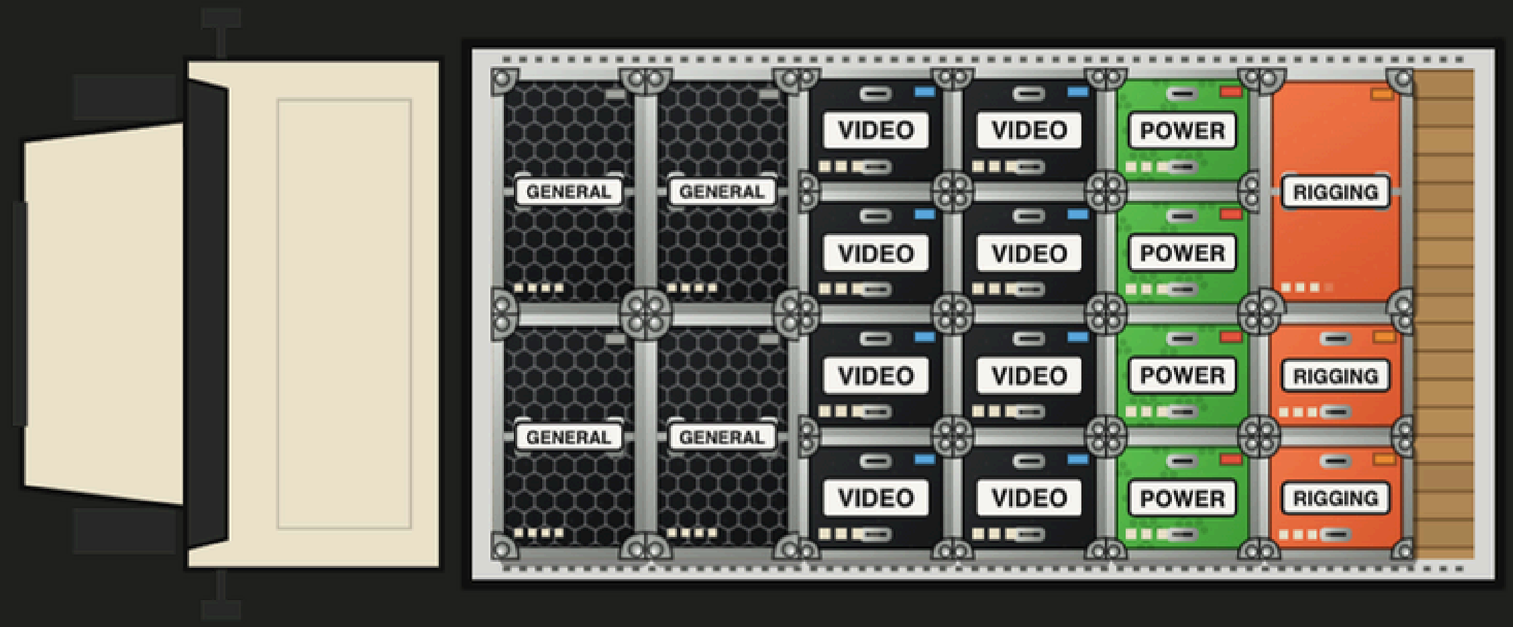
Pockets unlock at level 65. Up to 6" of space around the case is OK. A boxed-in case can't be picked out until the cases between it and the doors are gone.

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LAST ON, FIRST OFF

Tonight's pack is tomorrow's load-in: pack in reverse of the venue's order

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VENUE LOAD-IN ORDER

1. RIGGING: hung first
2. POWER: run next
3. VIDEO
4. GENERAL: last

The truck is packed in reverse: what the venue needs first goes on last, right by the doors.

ON A REAL TRUCK

Get the venue's load-in order before you pack. Truss and rigging ride by the doors. Unload from the top down, tier by tier from the doors, taking out bars and straps as you reach them.

IN THE GAME

Every level card shows the load-in order. Last on, first off starts at level 9, and levels 91–100 are unloading. Heavy still goes in the nose: the order works inside that.

THE CREW

One voice runs the load. In the game, four crew members coach you through it.



CREW CHIEF

Levels 1–30 · the floor



LEAD STAGEHAND

Levels 31–60 · stacking



RIGGER

Levels 61–90 · flipping



DOCK LEAD

Levels 91–100 · unloading

ON A REAL TRUCK

- One crew chief directs the load: one voice, no arguing in the truck.
- Stackers work inside the truck; pushers bring cases up the dock.
- Two people on every case going onto a stack.
- Safety meeting first, then no phones and no distractions on the load.

ACKNOWLEDGEMENT OF TRAINING

Truck Packology Mini Manual · to be signed by the Freelancer and kept by the Client

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By signing below, the undersigned freelance crew member (the "Freelancer") acknowledges, represents and agrees as follows:

1. Review. The Freelancer has received, read and understood each page of the Truck Packology Mini Manual (the "Manual") in its entirety, and has had the opportunity to ask questions about its contents.
2. Responsibilities. The Freelancer accepts the duties of a truck packer, including the safe loading, securing and unloading of equipment; the correct use of straps, load bars, caster cups, dock plates and pallet jacks; and compliance with the directions of the crew chief.
3. Truck Packology. The Freelancer has completed, or agrees to complete, the Truck Packology training game at games.aveducate.com/truck-packology.
4. Safety. The Freelancer will stop work and seek direction whenever a load, piece of equipment or working condition appears unsafe.
5. Educational purpose. The Manual and the game are educational tools only. They do not replace site-specific training, the instructions of the Client, or any applicable law, regulation or safety standard.

FREELANCER

Printed name

Signature

Date

CLIENT

Company

Authorized representative

Signature

Date

This acknowledgement is a record of training only. It does not create an employment relationship or replace any services agreement between the Freelancer and the Client.

APPROVED TRAINING PARTNERS

Labor companies that use this manual and Truck Packology to prepare their crews

AVEDUCATE.COM

The labor companies listed below have reviewed and approved the Truck Packology Mini Manual and the Truck Packology game as educational tools to help prepare their crews for the demands of truck loading.

COMPANY LOGO Approved by · Date	COMPANY LOGO Approved by · Date	COMPANY LOGO Approved by · Date	COMPANY LOGO Approved by · Date
COMPANY LOGO Approved by · Date	COMPANY LOGO Approved by · Date	COMPANY LOGO Approved by · Date	COMPANY LOGO Approved by · Date

Listing on this page means the company approved the manual and game for training use. It is not an endorsement of any individual crew member.

YOUR NEXT STEP

Build Your Reps. Earn Your Certificate.

101 levels. One final exam. A Truck Packing Fundamentals certificate from AV Educate.

SCAN TO START LEVEL 1

games.aveducate.com/truck-packology

THE BOOK	LEVELS 1–30	LEVELS 31–60	LEVELS 61–90	LEVELS 91–100	LEVEL 101	CERTIFIED
✓ Done. You are here.	The floor	Stacking	Expert packs	Unloading	Final exam	Truck Packing Fundamentals

The book is your agreement with the client. The game is the proof that you put in the reps before you get put on the event.

RULES & SOURCES

Every rule this book is built on, and who sets it

AVEDUCATE.COM

CARGO SECUREMENT · U.S. DOT, FMCSA

- 49 CFR Part 393, Subpart I (§393.100–136): protection against shifting and falling cargo
- §393.102: securement must hold 0.8 g forward, 0.5 g rearward and 0.5 g sideways
- §393.104: tiedowns in good shape: no knots, no damaged straps, edges protected
- §393.106(d): straps together rated for at least half the cargo's weight; wall to wall counts at full rating
- §393.108: working load limits and WLL markings

LIFTING · NIOSH & INTERNATIONAL

- NIOSH Revised Lifting Equation (Pub. 94-110): 51 lb load constant, reach and twist limits
- EN 1005-2 (European standard): team-lift factor, used for the max column
- UK HSE, Manual Handling Operations Regulations 1992 guidance (L23): team lifts at 2/3 and 1/2, used for the safer target
- Ergonomics guidance: stack by hand to shoulder height (60–65 in) and the 3:1 rule

DOCKS & STORAGE · U.S. DOL, OSHA

- 29 CFR 1910.26 Dockboards: rated for the load, secured in place, handholds, truck kept from moving
- 29 CFR 1910.178 Powered industrial trucks: forklift rules, brakes set and wheels chocked
- 29 CFR 1910.176(b) Secure storage: stacks stable against sliding or collapse
- OSH Act §5(a)(1) General Duty Clause: no single lift-weight limit; employers must prevent known hazards

INDUSTRY & MAKERS

- Freeman AV Standard 0024: AV crew practice on ramps and double-stacked cases
- ANSI MH30.2: portable dock leveling devices (dock plates)
- Maker ratings: strap WLL tags, load-bar capacity, dock-plate range, truck inside sizes
- Heavy-truck stability research: tall loaded trucks can roll near 0.35 g

IGNORANTIA JURIS NON EXCUSAT · Ignorance of the law excuses no one.

Courts presume everyone knows the law, so "I didn't know" is never a defense. If it were, anyone could claim it and no rule could be enforced. This manual is training material, not legal advice. These rules were checked in October 2026 and can change: always read the current text at ecfr.gov and osha.gov.